

A brewer by trade and a keen amateur scientist, British physicist James Prescott Joule investigated the nature of heat and the transfer of energy. He was renowned for experiments that proved heat was a form of energy and for confirming one of the most fundamental ideas in physics: the law of conservation of energy.

MILESTONES

BREWER TO SCIENTIST

Takes over the family brewery in 1837, which fuels his experiments into heat, pressure, and energy.

JOULE'S FIRST LAW

Establishes how much heat is generated by an electric current; presents findings in 1841.

ENERGY WORK

Results of an experiment he publishes in 1843 determine the mechanical equivalent of heat.

ROYAL SOCIETY

Accepts a fellowship to the Royal Society in 1850 and is awarded a Royal Medal 2 years later.

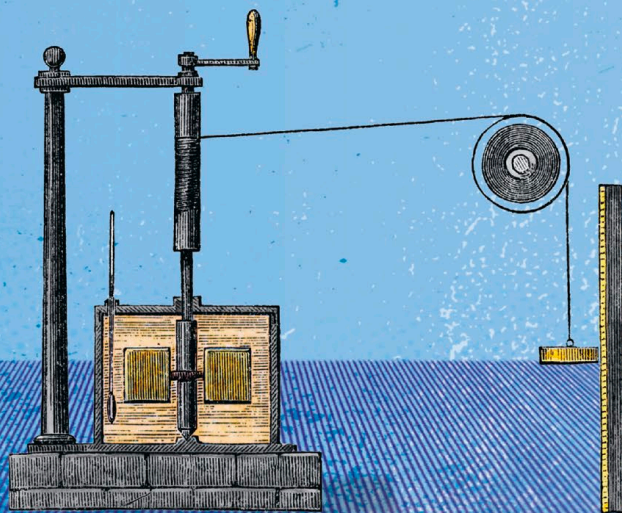
The owner of a successful brewery in northern England, James Prescott Joule fostered his interest in science by conducting experiments into the nature of heat in its laboratories. Although lacking formal training, he was a talented experimenter who made key discoveries during his career.

Experimental science

The results of Joule's early investigations into the relationship between electrical resistance and heat became known as Joule's first law. He then examined the direct relationship between mechanical work and heat, devising an experiment to mechanically turn a paddlewheel in water to raise its temperature. Joule concluded that the mechanical energy was converted into heat and argued that energy is never lost, but merely changes form. This groundbreaking theory became the foundation of the law of conservation of energy. He also discovered the Joule-Thomson effect with Lord Kelvin in 1852, which examined expansion and temperature changes in gases and led to pioneering work in refrigeration.

“Energy may be **converted** into **heat**; **heat** into energy.”

James Prescott Joule, 1847



Joule's most famous experiment used falling weights (right) to mechanically churn and heat water (left). He noted that the falling weights generated energy that converted into heat as the paddles created friction in the water.